**COURSE NAME: COMPUTER VISION**

**COURSE CODE: ITA0515**

**CO-2-ASSESSMENT TOOL 1: Analytical Problem Solving**

**Image Enhancement – Answers**

**1. Grey Level Transformation**

**(a) Purpose of Grey Level Transformation**

Grey level transformation is used to improve the visual quality of an image by changing the intensity values of pixels. It helps to:

- Enhance image contrast.
- Highlight important features.
- Brighten or darken an image.
- Prepare images for further processing.

**(b) Negative Transformation**

Given:

[
s = 255 - r
]

Pixel values:

[
r = [50,;100,;150,;200]
]

Calculations:

- (255 - 50 = 205)
- (255 - 100 = 155)
- (255 - 150 = 105)
- (255 - 200 = 55)

**Transformed values:**

$$
\boxed{[205,;155,;105,;55]}
$$

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**2. Log Transformation**

**(a) How Log Transformation Enhances Low-Intensity Pixels**

Log transformation expands the values of dark (low-intensity) pixels while compressing the values of bright (high-intensity) pixels. This improves the visibility of details in darker regions of an image without greatly affecting brighter areas.

**(b) Calculation**

Given:

$$
s = c\log(1+r), \quad c=10
$$

Using natural logarithm ($\ln$):

| (r) | Calculation | (s) |
|---|---|---|
| 0 | $10\ln(1)$ | **0.00** |
| 10 | $10\ln(11)$ | **23.98** |
| 50 | $10\ln(51)$ | **39.32** |
| 100 | $10\ln(101)$ | **46.15** |

**Final values:**

$$
\boxed{[0.00,;23.98,;39.32,;46.15]}
$$

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**3. Histogram Processing**

**(a) Role of Histogram Equalization**

Histogram equalization improves image contrast by redistributing the pixel intensity values over the entire available range. It makes dark regions brighter and bright regions more distinguishable, producing a clearer image.

**(b) Cumulative Distribution and Equalized Intensities**

Given histogram:

| **Intensity** | **Frequency** |
|---|---|
| 0 | 2 |
| 1 | 2 |
| 2 | 4 |
| 3 | 8 |

Total number of pixels:

$$N = 2+2+4+8 = 16$$

For a 3-bit image:

$$L = 2^3 = 8$$

Cumulative frequencies:

| **Intensity** | **Frequency** | **Cumulative Frequency** | **CDF** | **Equalized Intensity** **$(=\text{round}((L-1)\times CDF))$** |
|---|---|---|---|---|
| 0 | 2 | 2 | 2/16 = 0.125 | **1** |
| 1 | 2 | 4 | 4/16 = 0.25 | **2** |
| 2 | 4 | 8 | 8/16 = 0.50 | **4** |
| 3 | 8 | 16 | 16/16 = 1.00 | **7** |

**Equalized intensity values:**

$$\boxed{0\rightarrow1,\;1\rightarrow2,\;2\rightarrow4,\;3\rightarrow7}$$

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#### 4. Spatial Filtering (Smoothing)

##### (a) Purpose of Spatial Filtering for Smoothing

Spatial smoothing reduces noise and small intensity variations by replacing each pixel with the average of its neighboring pixels. This produces a smoother image and removes unwanted details.

**(b) Apply a 3 × 3 Averaging Filter**

Given matrix:

$$
\begin{matrix}
10 & 20 & 30\\
20 & 30 & 40\\
30 & 40 & 50
\end{matrix}
$$

Sum of all values:

$$
10+20+30+20+30+40+30+40+50 = 270
$$

Average:

$$
\frac{270}{9}=30
$$

**New center pixel value:**

$$
\boxed{30}
$$

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**5. Spatial Filtering (Sharpening)**

**(a) How Sharpening Enhances Image Details**

Sharpening increases the visibility of edges and fine details by emphasizing rapid changes in intensity. It makes images appear clearer and improves edge definition.

**(b) Apply the Laplacian Mask**

Laplacian mask:

[
\begin{bmatrix}
0 & -1 & 0\
-1 & 4 & -1\

0 & -1 & 0
\end{bmatrix}
]

Image matrix:

[
\begin{matrix}
10 & 10 & 10\
10 & 20 & 10\
10 & 10 & 10
\end{matrix}
]

Calculation at the center:

[
(4 \times 20) - (10+10+10+10)
]

[
=80-40
]

[
=40
]

**Output at the center:**

[
\boxed{40}
]

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**Final Answers**

1. **Negative Transformation: [205, 155, 105, 55]**
2. **Log Transformation: [0.00, 23.98, 39.32, 46.15]**
3. **Histogram Equalization:**

- Cumulative Frequencies: **2, 4, 8, 16**
- Equalized Intensities: **0→1, 1→2, 2→4, 3→7**

4. **3 × 3 Averaging Filter Output: 30**
5. **Laplacian Filter Output (Center Pixel): 40**